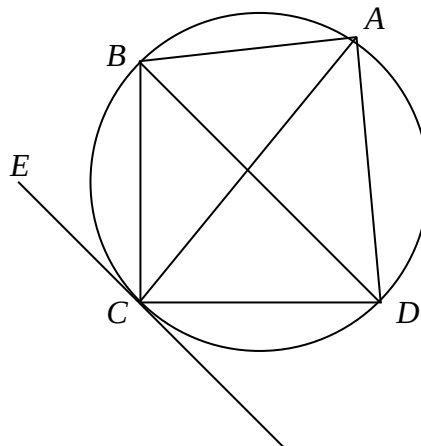


Question 1 12 Marks**Marks**

- (a) Evaluate the following definite integral $\int_{-2}^2 \frac{dx}{x^2 + 4}$ 2
- (b) Solve $\frac{5}{x-3} \geq 2$. 2
- (c) Show that $\int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$, where C is constant. 2
- (d) Find the general solution for $\cos 2\theta = \frac{\sqrt{3}}{2}$ 2
- (e) (i) Show that the derivative of $\frac{1+\sin x}{\cos x}$ is $\frac{1}{1-\sin x}$. 4
- (ii) Hence, deduce that $\int_0^{\pi/4} \frac{dx}{1-\sin x} = \sqrt{2}$

Question 2 12 Marks Start a new booklet

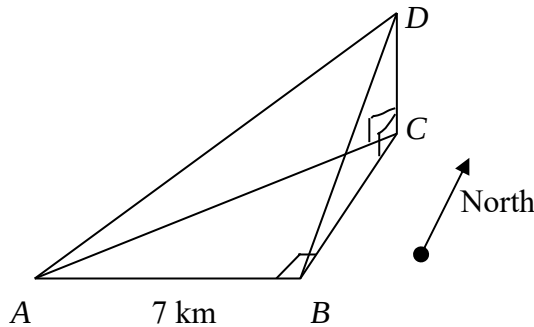
- (a) Let $f(x) = x^3 + 5x^2 + 17x - 10$. The equation $f(x) = 0$ has only one real root. 4
- (i) Show that the root lies between 0 and 2.
- (ii) Use one application of the 'halving the interval' method to find a smaller interval containing the root.
- (iii) Which end of the smaller interval found in part (ii) is closer to the root? Briefly justify your answer.
- (b) Evaluate $\int_{-1}^2 \frac{xdx}{\sqrt{3-x}}$ using the substitution $x = 3 - u$. 4
- (c) ABCD is a cyclic quadrilateral in which AC bisects $\angle DAB$. CE is the tangent to the circle at C. Prove $CE \parallel DB$. 4



Question 3 page 2.

Question 3 12 Marks Start a new booklet**Marks**

- (a) (i) Express $\sin \theta + \sqrt{3} \cos \theta$ in the form $R \sin(\theta + \phi)$. 4
- (ii) Hence, or otherwise, solve the equation $\sin \theta + \sqrt{3} \cos \theta = 1$ for values of θ between 0 and 2π .
- (b) Cadel notices that the angle of elevation of the top of a mountain due north is 14° . Upon riding 7 kilometres due west, he finds that the angle of elevation of the top of the mountain is 10° . How high is the mountain? Give your answer correct to the nearest metre. 4



$$\angle DBC = 14^\circ, \angle DAC = 10^\circ$$

- (c) (i) Show that $\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1$ 4
- (ii) Hence, show that $\frac{1}{1 + \sec x} = 1 - \frac{1}{2} \sec^2 \frac{x}{2}$.
- (iii) Use part (ii) to deduce that $\int_0^{\pi/2} \frac{dx}{1 + \sec x} = \frac{\pi}{2} - 1$.

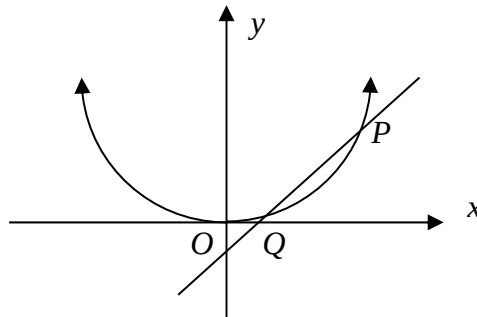
Question 4 page 3.

Question 4 12 Marks Start a new booklet

- (a) The sides of a cube are increasing at a rate of 2 cms^{-1} . Find at what rate the surface area is increasing when the sides are each 10 cm. 2
- (b) Prove by induction $9^{n+2} - 4^n$ is divisible by 5, for $n \geq 1$ 4
- (c) 100 grams of cane sugar in water is converted into dextrose at a rate which is proportional to the amount unconverted at any time. That is, if m grams are converted in t minutes, then $\frac{dm}{dt} = k(100 - m)$, where k is constant. 6
- (i) Show that $m = 100 + Ae^{-kt}$, where A is a constant, satisfies this equation.
- (ii) Find the value of A .
- (iii) If 40 grams are converted in the first 10 minutes, find how many grams are converted in the first 30 minutes.
- (iv) What is the limiting value of m as t increases indefinitely?

Question 5 12 Marks Start a new booklet

- (a) Solve the equation $6x^3 - 17x^2 - 5x + 6 = 0$, given that two of its roots have a product of -2 . 3
- (b) Find the values of a and b so that $x^4 + 4x^3 - x^2 + ax + b$ is divisible by $(x-2)(x+1)$. 3
- (c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. 6



- (i) Show that the equation of the chord PQ is $\frac{(p+q)x}{2} - y = apq$
- (ii) The line PQ passes through the point $(0, -a)$. Show that $pq = 1$.
- (iv) Hence, or otherwise, if S is the focus of the parabola, show that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$.

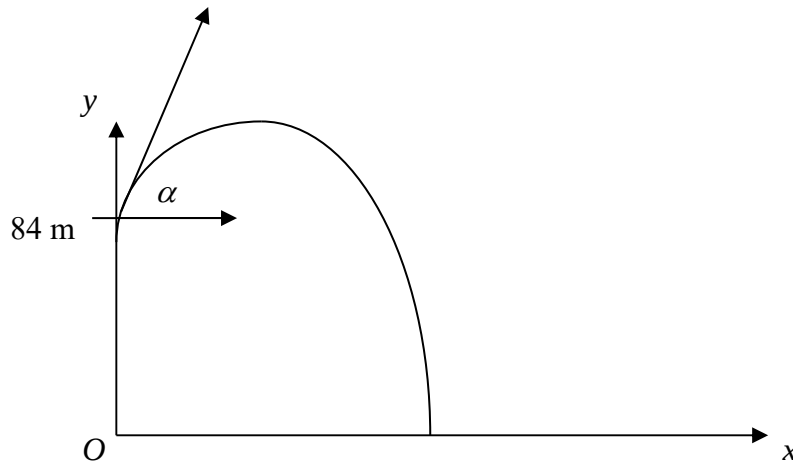
Question 6 12 Marks Start a new booklet

- (a) By considering the expansion of $(1+x)^{2n}$ in ascending powers of x show that ${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n} = 4^n$ 2
- (b) (i) Write the expansion of $(2+3x)^{20}$ in the form $\sum_{r=0}^{20} c_r x^r$, where c_r is the coefficient of x^r in the expansion. 4
- (ii) Show that $\frac{c_{r+1}}{c_r} = \frac{60-3r}{2r+2}$
- (iii) Hence, or otherwise, find the greatest coefficient in the expansion of $(2+3x)^{20}$. Leave your answer in index form.
- (c) Of a set of otherwise similar cards, ten are white, six are red and four are yellow. Three cards are taken at random. What is the probability that: 3
- (i) They are all different colours;
- (ii) They are the same colour?
- (d) (i) In how many ways can 2 boys and 1 girl be arranged in a row if the selection is made from 4 boys and 3 girls? 3
- (ii) In how many of these arrangements does a girl occupy the middle position?

Question 7 page 5.

Question 7 12 Marks Start a new booklet

- (a) A particle is projected with speed 40 ms^{-1} from the top of a cliff 84 metres high at an angle of elevation $\alpha = \tan^{-1} \frac{4}{3}$. Assume that the equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -10 \text{ ms}^{-2}$. 6



- (i) Derive the equations $x = 24t$ and $y = 84 + 32t - 5t^2$.
 - (ii) Hence or otherwise find the range on the horizontal plane through the foot of the cliff.
 - (iii) Find the speed of the body when it reaches this plane. Answer correct to two significant figures.
- (b) A particle moves in simple harmonic motion about $x = 0$ and its displacement x metres, at time t seconds, is given by $x = a \sin n(t + \alpha)$. The particle moves with a period of 16 seconds. It passes through the centre of motion when $t = 2$ seconds. Its velocity is 4 ms^{-1} when $t = 4$ seconds. 6

- (i) Show $\ddot{x} = -\frac{\pi^2}{64}x$.
- (ii) Find the maximum displacement.
- (iii) Find the speed of the particle when $t = 10$ seconds.

END OF EXAM

Question 1

$$(a) \int x \sin(x^2) dx$$

$$= -\frac{\cos(x^2)}{2} + C \quad \checkmark$$

$$(b) \int_0^{\frac{1}{2}} \cos^{-1} x dx = \int_0^{\frac{1}{2}} \cos^{-1} x \frac{d[x]}{dx} \cdot dx$$

$$= \left[x \cos^{-1} x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{-x}{\sqrt{1-x^2}} dx \quad \checkmark$$

$$= \frac{1}{2} \times \frac{\pi}{3} - \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}} \quad \checkmark$$

$$= \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \quad \checkmark$$

$$(c) (i) \text{ let } x \equiv a(x+3) + b(x-2)$$

$$\text{let } x=2 \quad \therefore a = \frac{2}{5} \quad \checkmark$$

$$x=-3 \quad \therefore b = \frac{3}{5} \quad \checkmark$$

$$\therefore \frac{x}{(x-2)(x+3)} = \frac{\frac{2}{5}}{x-2} + \frac{\frac{3}{5}}{x+3}$$

$$(ii) \quad \therefore \int \frac{x}{(x-2)(x+3)} dx = \frac{2}{5} \int \frac{1}{x-2} dx + \frac{3}{5} \int \frac{1}{x+3} dx \quad \checkmark$$

$$= \frac{2}{5} \ln|x-2| + \frac{3}{5} \ln|x+3| + C \quad \checkmark$$

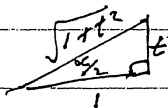
$$(d) \int \frac{x+4}{x^2-4x+13} dx = \frac{1}{2} \int \frac{2x-4}{x^2-4x+13} dx + \int \frac{6}{(x-2)^2+9} dx \quad \checkmark$$

$$= \frac{1}{2} \ln|x^2-4x+13| + 2 \tan^{-1} \frac{x-2}{3} + C \quad \checkmark$$

$$(e) \quad t = \tan \frac{x}{2}$$

$$\therefore x = 2 \tan^{-1} t$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$



$$\therefore \cos x = 2 \cos^2 \frac{x}{2} - 1$$

$$= \frac{1-t^2}{1+t^2}$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos x} = \int_0^1 \frac{\frac{2}{1+t^2}}{1 + \frac{1-t^2}{1+t^2}} dt \quad \checkmark$$

$$= \int_0^1 \frac{2}{1+t^2+1-t^2} dt \quad \checkmark$$

$$= \int_0^1 1 dt$$

$$= [t]_0^1$$

$$= 1 \quad \checkmark$$

Question 2

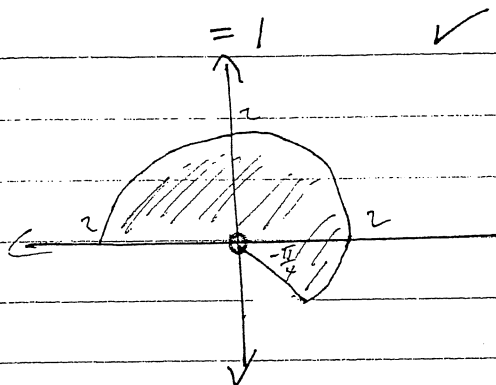
(a) $w = 1+i$, $z = 1-i\sqrt{3}$

(i) $w\bar{z} = (1+i)(1+i\sqrt{3})$
 $= (1-\sqrt{3}) + (1+\sqrt{3})i$ ✓

(ii) $\frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{1-i}{2}$
 $= \frac{1}{2} - \frac{1}{2}i$ ✓

(iii) $\frac{i[\operatorname{Re}(z) - z]}{g_m(z)} = \frac{i[1 - 1 + i\sqrt{3}]}{-\sqrt{3}}$ ✓

(b)



(c) $x^2 - 4x + (-4i) = 0$

$x = \frac{4 \pm \sqrt{16 - 4(1-4i)}}{2}$ ✓

$= \frac{4 \pm \sqrt{12 + 16i}}{2}$

$= 2 \pm \sqrt{3 + 4i}$

$\therefore x = 2 + (2+i)$

$= 4+i$

or $x = 2 - 2 - i$

$= -i$ ✓

Now let $\sqrt{3+4i} = x+iy$

$\therefore x^2 - y^2 = 3$

$2xy = 4 \therefore y = \frac{2}{x}$

$\therefore x^2 - \frac{4}{x^2} = 3$

$(x^2)^2 - 3(x^2) - 4 = 0$ ✓

$(x^2 - 4)(x^2 + 1) = 0$ ✗

$\therefore x = \pm 2 \therefore$ when $x = 2, y = 1$
 when $x = -2, y = -1$

(d)(i) $z = 1 + 2i + t(3 - 4i)$

$= 1 + 2i + 3t - 4ti$

$= (1+3t) + (2-4t)i$

$\therefore x = 1+3t \therefore (i)$

$y = 2-4t \therefore (ii)$ ✓

from (i) $t = \frac{x-1}{3}$

Sub into (ii) $y = 2 - \frac{4(x-1)}{3}$

$\therefore 3y = 6 - 4x + 4$

$\therefore 4x + 3y = 10$ ✓

(ii)

From $4x + 3y = 10$ ✓

$|z|$ minimum at

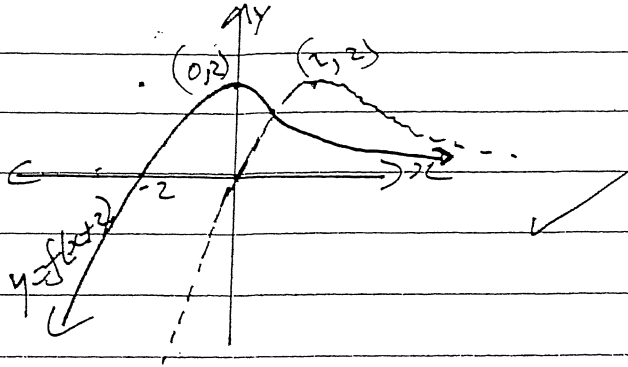
$d = \frac{|4(0) + 3(0) - 10|}{\sqrt{4^2 + 3^2}}$

$= \frac{|-10|}{5}$

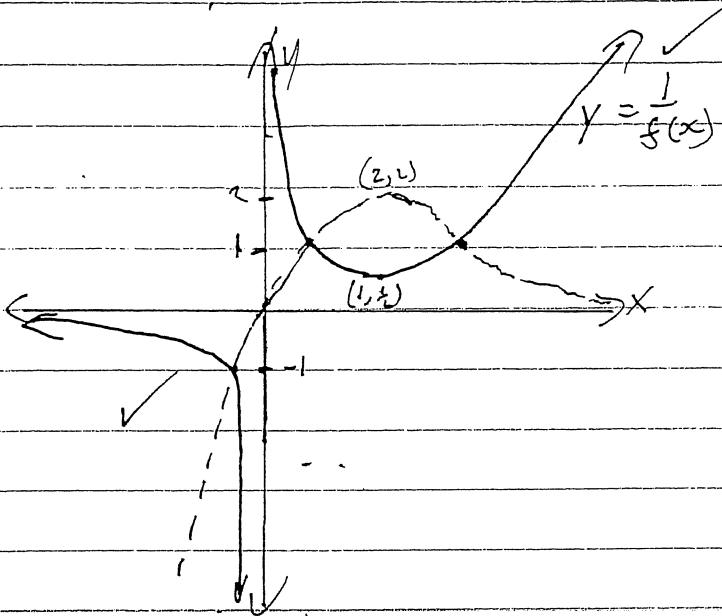
$= 2$ ✓

Question 3

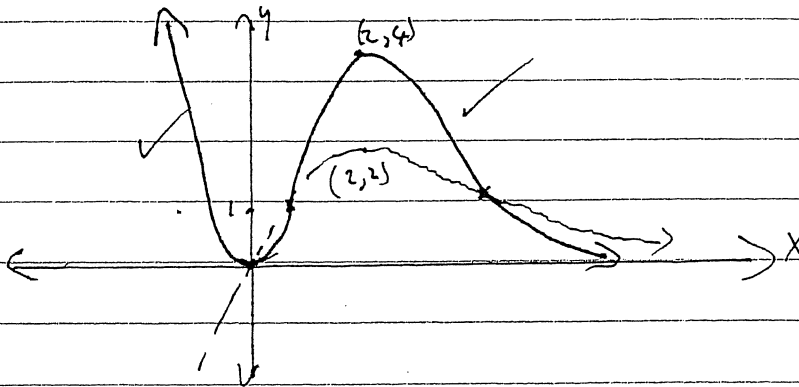
a) (i)



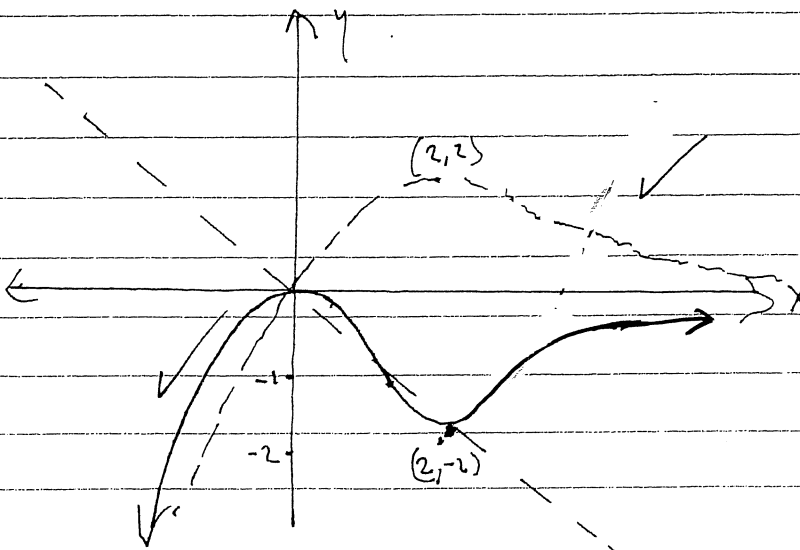
(ii)



(iii)



(iv)



Question 3 (Continued)

$$3(b)(i) \quad P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$$

$$P'(x) = 5x^4 - 12x^3 + 12x^2 - 8x + 3 \quad P'(1) = 5 - 12 + 12 - 8 + 3$$

$$P''(x) = 20x^3 - 36x^2 + 24x - 8$$

$$= 0$$

$$P''(1) = 20 - 36 + 24 - 8 = 0$$

$$\text{Now } P(1) = 1 - 3 + 4 - 4 + 3 - 1$$

$$= 0$$

$$\therefore P(1) = P'(1) = P''(1) = 0 \quad \checkmark$$

$\therefore$ triple root at $x=1$

1 Mark for
finding $P'(x)$ & $P''(x)$
1 Mark for
conclusion

$$(ii) \quad P(i) = i^5 - 3i^4 + 4i^3 + 4i^2 + 3i - 1$$

$$\text{as } i^2 = -1 \quad = i - 3 + 4i + 4 + 3i - 1 \quad \checkmark$$

$$= 0$$

$$(iii) \quad \text{if } x=i \text{ is a root } \therefore P(-i) = 0 \quad \checkmark$$

$\therefore$ Roots are $1, i, -i$ $\checkmark$

$$(c) \quad x^3 - 2x^2 - 5x - 1 = 0$$

Let new equation have roots in the form

$$x = \frac{1}{\alpha^2}$$

$$\therefore \alpha^2 = \frac{1}{x}$$

$$\alpha = \pm \frac{1}{\sqrt{x}} \quad \checkmark$$

$$\text{as } \alpha \text{ is a root of } x^3 - 2x^2 - 5x - 1 = 0$$

$$\therefore \left(\pm \frac{1}{\sqrt{x}}\right)^3 - 2\left(\pm \frac{1}{\sqrt{x}}\right)^2 - 5\left(\pm \frac{1}{\sqrt{x}}\right) - 1 = 0$$

$$\pm \frac{1}{\sqrt{x}} \left(\frac{1}{x} - 5\right) = \frac{2}{x} + 1$$

$$\therefore \frac{1}{x} \left(\frac{1}{x} - 5\right)^2 = \left(\frac{2}{x} + 1\right)^2 \quad \checkmark$$

$$\therefore \frac{1}{x} \left(\frac{1}{x^2} - \frac{10}{x} + 25\right) = \frac{4}{x^2} + \frac{4}{x} + 1$$

$$\frac{1}{x^3} - \frac{10}{x^2} + \frac{25}{x} = \frac{4}{x^2} + \frac{4}{x} + 1$$

$$\therefore 1 - 10x + 25x^2 = 4x + 4x^2 + x^3$$

$$\therefore x^3 - 21x^2 + 14x - 1 = 0 \quad \text{has roots } \frac{1}{\alpha^2}, \frac{1}{\alpha^2}, \frac{1}{\alpha^2} \quad \checkmark$$

Note students may
get the same result using:

$$x^3 - (\text{Sum})x^2 + (\text{Sum of products})x - \text{Product} = 0$$

Question 4

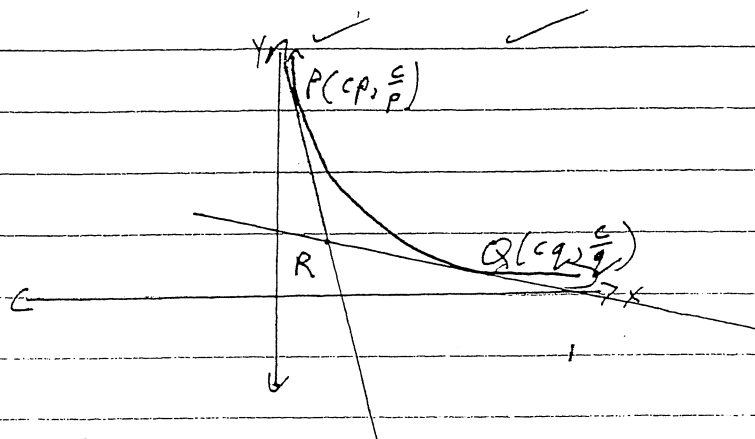
(A) (i) Circle $5 - K = K - 3$ ✓

$K = 4$ ✓

(ii) Hyperbola $5 - K < 0$ or $K - 3 < 0$, but not both

$\therefore K > 5$ or $K < 3$

(b)



(i) $xy = c^2$

$x \frac{dy}{dx} + y = 0$

$\frac{dy}{dx} = -\frac{y}{x}$

at $T(ct, \frac{c}{t})$ $\frac{dy}{dx} = -\frac{c/t}{ct}$
 $= -\frac{1}{t^2}$

$\therefore y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$ ✓

$t^2y - ct = -x + ct$

$\therefore x + t^2y - 2ct = 0$

(ii) at P the tangent is $x + p^2y - 2cp = 0$ --- (1)

$x + q^2y - 2cq = 0$ --- (2)

$\therefore$ (1) - (2) gives $(p^2 - q^2)y = 2c(p - q)$

$y = \frac{2c}{p + q}$ ✓

Sub into (1)

$\therefore x + p^2 \left(\frac{2c}{p + q} \right) = 2cp$

$x = 2cp - \frac{2cp^2}{p + q}$

$= \frac{2cp^2 + 2cpq - 2cp^2}{p + q}$

$= \frac{2cpq}{p + q}$ ✓

$\therefore R \left[\frac{2cpq}{p + q}, \frac{2c}{p + q} \right]$

Question 4 (Continued)

Q4(b)

(iii) If $xg + y^2 = 2c \therefore R\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ satisfies it.

$$\begin{aligned} \text{LHS} &= \frac{2cpq}{p+q} \cdot \frac{2c}{p+q} + \left(\frac{2c}{p+q}\right)^2 \quad \checkmark \quad \text{[ALTERNATE SOLUTION ATTACHED]} \\ &= \frac{4c^2pq}{(p+q)^2} + \frac{4c^2}{(p+q)^2} \\ &= \frac{4c^2(pq+1)}{p^2+q^2+2pq} \\ &= \frac{4c^2(pq+1)}{2+2pq} \quad \text{given } p^2+q^2=2 \quad \checkmark \\ &= \frac{4c^2(pq+1)}{2(pq+1)} \\ &= 2c^2 \quad \checkmark \\ &= \text{RHS} \quad \neq \end{aligned}$$

4(c) $P(a \cos \theta, b \sin \theta)$, $Q(a \cos \phi, b \sin \phi)$ on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$(i) \quad M_{PQ} = \frac{b(\sin \phi - \sin \theta)}{a(\cos \phi - \cos \theta)} \quad \checkmark$$

$$\begin{aligned} \therefore \text{Equation of chord: } y - b \sin \theta &= \frac{b(\sin \phi - \sin \theta)}{a(\cos \phi - \cos \theta)} (x - a \cos \theta) \quad \checkmark \\ \therefore ay(\cos \phi - \cos \theta) &= ab \sin \theta (\cos \phi - \cos \theta) = b(\sin \phi - \sin \theta)x - ab \cos \theta (\sin \phi - \sin \theta) \end{aligned}$$

(ii) Focal chord passes through $(ae, 0)$ $\checkmark$

$$\begin{aligned} \therefore a b e (\sin \phi - \sin \theta) &= a b \cos \theta (\sin \phi - \sin \theta) - a b \sin \theta (\cos \phi - \cos \theta) \quad \checkmark \\ e (\sin \phi - \sin \theta) &= \cos \theta \sin \phi - \cos \theta \sin \theta - \sin \theta \cos \phi + \sin \theta \cos \theta \quad \checkmark \\ &= \sin(\phi - \theta) \\ \therefore e &= \frac{\sin(\phi - \theta)}{\sin \phi - \sin \theta} \quad \checkmark \end{aligned}$$

Q 4(b)(ii) ALTERNATE SOLUTION

$$\text{Given } p^2 + q^2 = 2$$

$$\therefore (p+q)^2 - 2pq = 2$$

$$\therefore pq = \frac{(p+q)^2 - 2}{2} \quad *$$

$$\text{Now R } \left[\frac{2cpq}{p+q}, \frac{2c}{p+q} \right]$$

$$\therefore y = \frac{2c}{p+q}$$

$$\therefore p+q = \frac{2c}{y}$$

$$\begin{aligned} \text{from } * \quad pq &= \frac{\left(\frac{2c}{y}\right)^2 - 2}{2} \\ &= \frac{4c^2 - 2y^2}{2y^2} \\ &= \frac{2c^2 - y^2}{y^2} \end{aligned}$$

$$\text{Now } x = \frac{2cpq}{p+q}$$

$$\therefore x = \frac{2c \times \frac{2c^2 - y^2}{y^2}}{\frac{2c}{y}}$$

$$\therefore x = \frac{2c^2 - y^2}{y}$$

$$xy = 2c^2 - y^2$$

$$\therefore xy + y^2 = 2c^2$$

Question 5

$$(a) N(\text{Sample}) = \frac{{}^{12}C_5 \cdot {}^7C_5}{2}$$

$$= 8316 \quad \checkmark$$

$$N(\text{Event}) = \frac{{}^3C_3 \times {}^9C_2 \times {}^7C_2}{8316} \quad \checkmark$$

$$= \frac{72}{792}$$

$$= \frac{1}{11} \quad \checkmark$$

3C_3 - triplets together

9C_2 - other two members of their team.

7C_2 - second team

$$(b) y = 4x - x^2$$

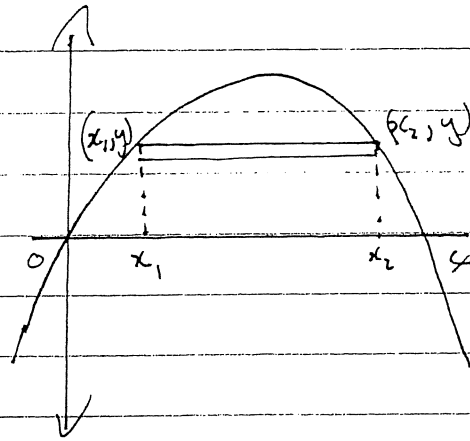
$$(i) \therefore x^2 - 4x + y = 0$$

$$x = \frac{4 \pm \sqrt{16 - 4y}}{2} \quad \checkmark$$

$$= 2 \pm \sqrt{4 - y}$$

$$\therefore x_2 = 2 + \sqrt{4 - y} \quad \checkmark$$

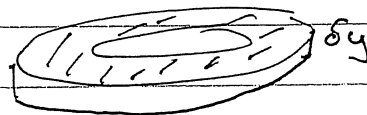
$$x_1 = 2 - \sqrt{4 - y}$$



(i) Max value of $y = 4 \quad \checkmark$ (By inspection)

(ii) Using slices perpendicular to y -axis

$\therefore$ Disc



$$\text{Volume of disc} = \pi (x_2^2 - x_1^2) \delta y \quad \checkmark$$

$$\text{Volume} = \lim_{\delta y \rightarrow 0} \sum_{y=0}^{y=4} \pi (x_2^2 - x_1^2) \delta y \quad \checkmark$$

$$= \pi \int_0^4 (x_2 - x_1)(x_2 + x_1) dy$$

$$= \pi \int_0^4 \sqrt{4 - y} dy \quad \checkmark$$

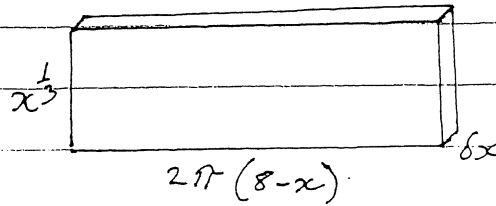
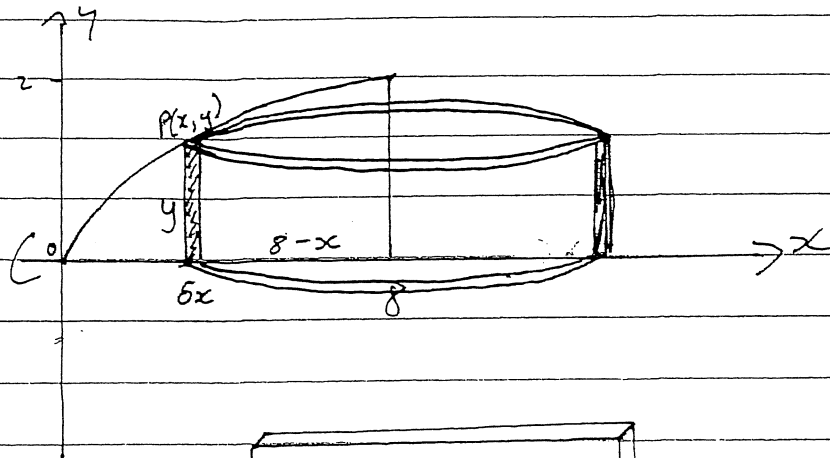
$$= \pi \left[-\frac{2}{3} (4 - y)^{3/2} \right]_0^4$$

$$= \frac{8\pi}{3} \cdot \frac{2}{3} (0 - 8)$$

$$= \frac{128\pi}{3} \text{ units}^3 \quad \checkmark$$

Question 5 (continued)

(c)



$\therefore \text{Volume of shell } \delta V = 2\pi(8-x)x^{\frac{1}{3}}\delta x$

$\therefore \text{Volume} = \lim_{\delta x \rightarrow 0} \sum_{x=0}^8 2\pi(8-x)x^{\frac{1}{3}}\delta x \quad \checkmark$

$= 2\pi \int_0^8 (8x^{\frac{1}{3}} - x^{\frac{4}{3}}) dx$

$= 2\pi \left[6x^{\frac{4}{3}} - \frac{3}{7}x^{\frac{7}{3}} \right]_0^8 \quad \checkmark$

$= 2\pi \left(6 \times 16 - \frac{3}{7} \times 128 \right)$

$= 2\pi \left(96 - \frac{384}{7} \right)$

$= \frac{576}{7}\pi \text{ units}^3 \quad \checkmark$

Question 6

(a) Given $z^n - \frac{1}{z^n} = 2i \sin \theta$ and $(z - \frac{1}{z})^5 = z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$

(i) $z - \frac{1}{z} = 2i \sin \theta$

Now $(z - \frac{1}{z})^5 = z^5 - \frac{1}{z^5} - 5(z^3 - \frac{1}{z^3}) + 10(z - \frac{1}{z})$ ✓

$\therefore (2i \sin \theta)^5 = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$ ✓

$32i \sin^5 \theta = 2i (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$

$\therefore \sin^5 \theta = \frac{1}{16} (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$ ✓

(ii) Now from (i)

$16 \sin^5 \theta = \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta$

$\therefore$ if $16 \sin^5 \theta = \sin 5\theta$

$\therefore -5 \sin 3\theta + 10 \sin \theta = 0$ ✓

$\therefore 2 \sin \theta = \sin 3\theta$

$= \sin(2\theta + \theta)$

$= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta$

$= 2 \sin \theta \cos^2 \theta + (1 - 2 \sin^2 \theta) \sin \theta$

$= 2 \sin \theta (1 - \sin^2 \theta) + \sin \theta - 2 \sin^3 \theta$

$= 3 \sin \theta - 4 \sin^3 \theta$

$\therefore 4 \sin^3 \theta - \sin \theta = 0$ ✓

$\therefore \sin \theta (4 \sin^2 \theta - 1) = 0$

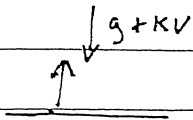
$\therefore \sin \theta (2 \sin \theta - 1)(2 \sin \theta + 1) = 0$

$\therefore \sin \theta = 0$ or $\sin \theta = \frac{1}{2}$ or $\sin \theta = -\frac{1}{2}$ ✓

$\therefore \theta = n\pi$ or $n\pi + (-1)^{\frac{\pi}{6}}$ or $n\pi + (-1)^{\frac{-\pi}{6}}$ ✓
 $2k\pi + \frac{\pi}{6}$ $(2k+1)\frac{-\pi}{6}$

Question 6 (continued)

(b) (i)



$$\dot{x} = -g - Kv$$

$$\therefore \frac{dv}{dt} = -g - Kv \quad \checkmark$$

$$\therefore \frac{dt}{dv} = \frac{1}{g + Kv}$$

$$t = -\frac{1}{K} \int_u^v \frac{K}{g + Kv} dv$$

$$= -\frac{1}{K} \left[\ln(g + Kv) \right]_u^v$$

$$= -\frac{1}{K} \ln\left(\frac{g + Kv}{g + Ku}\right) \quad \checkmark$$

$$\therefore -Kt = \ln\left(\frac{g + Kv}{g + Ku}\right)$$

$$\therefore g + Kv = e^{-Kt}(g + Ku)$$

$$\therefore v = \frac{g + Ku}{K} e^{-Kt} - \frac{g}{K} \quad \checkmark$$

$$\therefore x = \int_0^t \left[\frac{g + Ku}{K} e^{-Kt} - \frac{g}{K} \right] dt \quad \checkmark$$

$$= -\frac{(g + Ku)}{K^2} e^{-Kt} - \frac{g}{K} t + \frac{g + Ku}{K^2}$$

$$= \frac{g + Ku}{K^2} (1 - e^{-Kt}) - \frac{gt}{K} \quad \checkmark$$

(ii) Now at $t = T$

$$\frac{g + Ku}{K^2} (1 - e^{-KT}) - \frac{gT}{K} = h + \frac{g}{K^2} (1 - e^{-KT}) - \frac{gT}{K} \quad \checkmark$$

$$\therefore (1 - e^{-KT}) \left[\frac{g + Ku}{K^2} - \frac{g}{K^2} \right] = h$$

$$1 - e^{-KT} = \frac{hK}{u}$$

$$e^{-KT} = \frac{u - hK}{u} \quad \checkmark$$

$$\therefore -KT = \ln\left(\frac{u - hK}{u}\right)$$

$$T = \frac{1}{K} \ln\left(\frac{u}{u - hK}\right) \quad \checkmark$$

Question 7

$$\begin{aligned} (a) (i) \int_0^a f(x) dx & \quad \text{let } u = a - x \\ &= \int_a^0 -f(a-u) du \quad \begin{array}{l} x = a - u \\ \frac{dx}{du} = -1 \end{array} \quad \begin{array}{l} x=0, u=a \\ x=a, u=0 \end{array} \\ &= \int_0^a f(a-u) du \\ &= \int_0^a f(a-x) dx \quad \checkmark \end{aligned}$$

$$(ii) \int_0^{\pi} x \cos^2 x dx = \int_0^{\pi} (\pi - x) \cos^2(\pi - x) dx$$

$$\int_0^{\pi} x \cos^2 x dx = \int_0^{\pi} (\pi \cos^2 x - x \cos^2 x) dx \quad \checkmark$$

$$\therefore 2 \int_0^{\pi} x \cos^2 x dx = \pi \int_0^{\pi} \cos^2 x dx \quad \checkmark$$

$$\therefore 2 \int_0^{\pi} x \cos^2 x dx = \frac{\pi}{2} \int_0^{\pi} (1 + \cos 2x) dx \quad \cos^2 x = \frac{1}{2}(\cos 2x + 1)$$

$$\int_0^{\pi} x \cos^2 x dx = \frac{\pi}{4} \left[x + \frac{1}{2} \sin 2x \right]_0^{\pi} \quad \checkmark$$

$$= \frac{\pi}{4} (\pi - 0)$$

$$= \frac{\pi^2}{4}$$

Question 7 (continued)

$$(b)(i) (\cos \theta + i \sin \theta)^n = (\cos \theta + i \sin \theta)^n$$

$$= \cos^n \theta + \binom{n}{1} \cos^{n-1} \theta (i \sin \theta) + \binom{n}{2} \cos^{n-2} \theta (i \sin \theta)^2 + \binom{n}{3} \cos^{n-3} \theta (i \sin \theta)^3 + \dots + \binom{n}{n} \cos^0 \theta (i \sin \theta)^n$$

$$= \cos^n \theta - \binom{n}{2} \cos^{n-2} \theta \sin^2 \theta + \binom{n}{4} \cos^{n-4} \theta \sin^4 \theta - \dots + i \left[\binom{n}{1} \cos^{n-1} \theta \sin \theta - \binom{n}{3} \cos^{n-3} \theta \sin^3 \theta + \dots \right]$$

as $i^2 = -1, i^3 = -i, i^4 = 1, \dots$

$\therefore$ Equating the real and imaginary parts of both sides

$$(1) \cos n\theta = \cos^n \theta - \binom{n}{2} \cos^{n-2} \theta \sin^2 \theta + \binom{n}{4} \cos^{n-4} \theta \sin^4 \theta - \dots$$

$$(2) \sin n\theta = \binom{n}{1} \cos^{n-1} \theta \sin \theta - \binom{n}{3} \cos^{n-3} \theta \sin^3 \theta + \dots$$

(ii) From (1) & (2)

$$\cos n\theta = \cos^n \theta \left[1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots \right]$$

$$= \cos^n \theta \left[1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots \right] \quad \text{--- (3)}$$

$$\sin n\theta = \cos^n \theta \left[\binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \dots \right] \quad \text{--- (4)}$$

$$\therefore (4) \div (3)$$

$$\therefore \tan n\theta = \frac{\binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \dots}{1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots}$$

(c) (i) Number of points of intersection = ${}^6C_2 = 15$ There are 6 lines with 5 points of intersection on each

(ii) $P(\text{four of these pts do not all lie on one of the given lines})$
 $= 1 - P(\text{four of these pts chosen at random all lie on one of the given lines})$

$$= 1 - \frac{6 \times {}^5C_4}{{}^{15}C_4} = \frac{89}{91}$$

Question 8

$$(a) \quad \cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y$$

$$\therefore \cos(x+y) - \cos(x-y) = -2 \sin x \sin y$$

$$\text{let } S = x+y$$

$$T = x-y$$

$$\therefore x = \frac{S+T}{2}, y = \frac{S-T}{2} \quad \therefore \cos S - \cos T = -2 \sin\left(\frac{S+T}{2}\right) \sin\left(\frac{S-T}{2}\right)$$

$$(b) \quad I_n = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2nx}{\sin 2x} dx$$

$$(i) \quad I_1 = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{\sin 2x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{2 \sin^2 x}{2 \sin x \cos x} dx \quad \text{as } \cos 2x = 1 - 2 \sin^2 x$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} dx \quad \checkmark$$

$$= - \left[\ln |\cos x| \right]_0^{\frac{\pi}{4}}$$

$$= - \ln \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} \ln 2 \quad \checkmark$$

$$(ii) \quad I_{2r+1} - I_{2r-1} = \int_0^{\frac{\pi}{4}} \frac{1 - \cos(4xr+2x)}{\sin 2x} - \frac{1 - \cos(4xr-2x)}{\sin 2x} dx \quad \checkmark$$

$$= \int_0^{\frac{\pi}{4}} \frac{\cos(4xr-2x) - \cos(4xr+2x)}{\sin 2x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{-2 \sin 4xr \cdot \sin(-2x)}{\sin 2x} dx \quad \text{From (a)} \quad \checkmark$$

$$= 2 \int_0^{\frac{\pi}{4}} \sin 4xr \cdot dx$$

$$= - \frac{1}{4r} x^2 \left[\cos 4xr \right]_0^{\frac{\pi}{4}}$$

$$= - \frac{1}{2r} \left[\cos r\pi - 1 \right]$$

$$= - \frac{1}{2r} \left[(-1)^r - 1 \right] = \frac{1 - (-1)^r}{2r} \quad \checkmark$$

Question 8 (continued)

$$(b) (iii) \text{ as } I_{2r+1} - I_{2r-1} = \frac{1 - (-1)^r}{2r} *$$

$$\therefore I_9 = I_{2 \times 4 + 1}$$

$$\therefore I_9 - I_7 = 0 \quad \text{from } *$$

$$\therefore I_9 = I_7$$

$$I_7 = I_{2 \times 3 + 1}$$

$$\therefore I_7 - I_5 = \frac{1+1}{6}$$

$$\therefore I_7 = \frac{1}{3} + I_5 \quad \checkmark$$

$$\text{Now } I_5 = I_{2 \times 2 + 1}$$

$$\therefore I_5 - I_3 = 0$$

$$\therefore I_5 = I_3$$

$$I_3 = I_{2 \times 1 + 1}$$

$$\therefore I_3 - I_1 = \frac{2}{2}$$

$$= 1 \quad \checkmark$$

$$I_3 = 1 + I_1$$

$$= 1 + \frac{1}{2} \ln 2 \quad \text{from (i)}$$

$$\therefore I_9 = \frac{1}{3} + 1 + \frac{1}{2} \ln 2 \quad \checkmark$$

